

Chapter 9 exercises — Words and Mappings

63 exercises, extracted from the textbook PDF.

Caveats. (1) Statements are auto-extracted; `fi` / `fl` ligatures dropped by the PDF text layer were repaired by a cleanup pass (a rare odd gap may remain). (2) A $\triangle$ marks exercises whose statement includes a **displayed formula** that did not survive extraction — read those in the book at the cited page. The book PDF is always authoritative.

#	p.	Exercise
9.1	480	How many different occupancy distributions are there for N unlabelled balls in a sequence of M urns? For example, if we denote the desired quantity by C_M^N , we have $C_2^N = N + 1$ because there is one configuration with k balls in the first urn and $N - k$ balls in the second urn for each k from 0 to N .
9.2	484	Give a table like Table 9.2 for three balls in four urns.
9.3	484	Give tables like Tables 9.2 and 9.4 for two urns.
9.4	484	Give necessary and sufficient conditions on N and M for the average number of empty urns to equal the average minimum urn occupancy.
9.5	488	For $M = 365$, how many people are needed to be 99% sure that two have the same birthday?
9.6	488	Estimate the variance associated with the birthday distribution that is given in eorem 9.1, and explain the apparent discrepancy concerning the asymptotic values of the mean and the median.
9.7	490	Find all the 2-collections and 3-collections in Table 9.2, then compute $P_2(z)$ and $P_3(z)$ and check the coefficients of z^4 .
9.8	491	Expand the PGF by partial fractions to show that the probability that the N th ball fills the last empty urn can also be expressed as the alternating sum $(\sum_{j=0}^{M-1} (-1)^j \binom{M-1}{j} \frac{1}{M-j})$.
9.9	491	Give an expression for the probability that collecting at least N boxes gives a full collection of M coupons.
9.10	493	Consider the “largest” construction among M -surjections: given an M -surjection of length N , consider the $(M - 1)$ -surjection formed by removing all occurrences of M . Find the EGF for surjections using this construction.
9.11	493	Write a program that takes N and M as parameters and prints out all M -surjections of length N whenever the number of such objects is less than 1000.
9.12	493	Expand $(ez - 1)^M$ by the binomial theorem to show that $N![z^N]F_M(z) =$
9.13	493	Show that the number of partitions of N elements into nonempty subsets is $z^N N![z^N]e^z - 1$.
9.14	493	Show that $N![z^N]e^z - 1 = z$

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9.15	493	Prove that the bivariate EGF for the Stirling numbers of the second kind is $\exp(u(ez - 1))$.
9.16	493	Applying the “largest” construction to find the number of M -words of length N might lead to the recurrence $F_N M =$
9.17	497	Find the EGF for M -words with all letter frequencies even.
9.18	497	Prove that the number of ways to distribute M (k + 1) balls among M urns with all urns having $> k$ balls is equal to the number of ways to distribute M (k + 1) – 1 balls among M urns with all urns having $< (k + 2)$ balls, for all $k \geq 0$. (See Table 9.5.) Give an explicit formula for this quantity as a quotient of factorials.
9.19	497	What is the expected number of balls thrown in N urns before the second collision occurs? Assume that “collision” here means the event “ball falling into nonempty urn.”
9.20	497	What is the expected number of balls thrown in N urns before the second collision occurs, when we assume that “collision” means the event “ball falling into urn with exactly one ball in it?”
9.21	497	Give an explicit expression for the number of M -words with no three occurrences of the same letter.
9.22	497	Give a plot like Figure 9.3 for the probability that three people have the same birthday.
9.23	497	For M = 365, how many people are needed to be 50% sure that three have the same birthday? Four?
9.24	499	What is the average number of balls thrown into M urns before each urn is filled at least twice?
9.25	499	Derive an expression for the exponential CGF for the expected minimal occupancy when N balls are distributed into M urns. Tabulate the values for M and N less than 20.
9.26	500	What is the average number of blocks of contiguous equal elements in a random word?
9.27	500	Analyze “rises” and “runs” in words (cf. §7.1).
9.28	505	What is the probability that one urn will get all the balls when 100 balls are randomly distributed among 100 urns?

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9.29	505	What is the probability that each urn will get one ball when 100 balls are randomly distributed among 100 urns?
9.30	505	What is the standard deviation for the average number of empty urns?
9.31	505	What is the probability that each urn will contain an even number of balls when N balls are distributed among M urns?
9.32	505	Prove that $[M]$
9.33	509	For 1000 keys, which value of M will make hashing with separate chaining access fewer keys than a binary tree search? For 1 million keys?
9.34	509	Find the standard deviation of the number of comparisons required for a successful search in hashing with separate chaining.
9.35	509	Determine the average and standard deviation of the number of comparisons used for a search when the lists in the table are kept in sorted order (so that a search can be cut short when a key larger than the search key is found).
9.36	509	[Broder and Karlin] Analyze the following variant of Program 9.1: compute two hash functions and put the key on the shorter of the two lists.
9.37	518	Find $[z^n] e^{\alpha C(z)}$ where $C(z)$ is the Cayley function (see the discussion at the end of §6.14 and in §9.7 in this chapter).
9.38	518	("Abel's binomial theorem.") Use the result of the previous exercise and the identity $e^{(\alpha+\beta)C(z)} = e^{\alpha C(z)} e^{\beta C(z)}$ to prove that $(\alpha + \beta)(n + \alpha + \beta)$
9.39	518	How many keys can be inserted into a linear probing table of size M before the average search cost gets to be greater than $\ln N$?
9.40	518	Compute the exact cost of an unsuccessful search using linear probing for a full table.
9.41	518	Give an explicit representation for the EGF for the cost of an unsuccessful search.
9.42	518	Use the symbolic method to derive the EGF of the number of probes required by linear probing in a successful search, for fixed M . *
9.43	520	Find the exponential BGF for the image cardinality distribution.
9.44	520	Use a combinatorial argument to find the exponential BGF for the image cardinality distribution.

#	p.	Exercise
9.45	520	Give a recurrence relationship for the number of mappings of size N with k different integers in the image, and use that to obtain a table of values for $N < 20$.
9.46	520	Give an explicit expression for the number of M -words of length N having k different letters.
9.47	522	Prove that every random mapping must have at least one cycle.
9.48	522	Explore properties of the random mappings defined by $f(i) \equiv 1 + (i^2 + 1) \bmod N$ for $N = 100, 1000, 10,000$, and primes near these values.
9.49	527	Show that the analysis of the rho length of a random point in a random mapping is equivalent to that for the birthday problem.
9.50	531	Which N -mappings have maximal and minimal rho length? Tree path length?
9.51	531	Write a program to find the rho length and tree path length of a random mapping. Generate 1000 random mappings for N as large as you can and compute the average number of cycles, rho length, and tree path length.
9.52	531	Write a program to find the rho length and tree path length of a random mapping without using any extra storage.
9.53	531	A mapping with no repeated integers is a permutation. Give an efficient algorithm for determining whether a mapping is a tree.
9.54	531	Compute the average size of the largest component in a random N mapping, for all $N < 10$.
9.55	531	$\triangle$ Prove
9.56	531	What is the average number of different integers in the image when a random mapping is iterated twice?
9.57	531	Consider the $N \times N$ tree-cycle structures that correspond to all the N mappings. How many of these are different when considered as unlabelled, unordered objects, for $N \leq 7$? (These are called random mapping patterns.)
9.58	531	Describe the graph structure of partial mappings, where the image of a point may be undefined. Set up the corresponding EGF equations and check that the number of partial mappings of size N is $(N + 1)N$.
9.59	532	Analyze "path length" in sequences of $2N$ random integers in the range 1 to N .

#	p.	Exercise
9.60	532	Generate 100 random mappings of size 10, 100, and 1000 and empirically verify the statistics given in Table 9.13.
9.61	533	Use Floyd's method to test the random number generators on your machine for short cycles.
9.62	533	Use Floyd's algorithm to test the middle square random number generator.
9.63	533	Use Floyd's method to estimate the rho length associated with various starting values, c, and N for the function $f(x) = (x^2 + c) \bmod N$.